

YIMIAN LIU

AI Engineering | Machine Learning & Trading System

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EXPERIENCE

Guzman Energy

Denver, CO

Operations Research Engineer (AI & Platform Tech Lead)

Jan 2026 – Present

- Led a three-person AI-native team build Guzman's core internal trading, AI and data platform from scratch in under two years: 32+ production apps, 71% of employees active weekly, \$1M daily trading volume, and 5B+ system events per day.
- Architected, built and owned a fully-automated production SPP trading system that integrates forecast with an innovative trader-in-the-loop formula-based strategy design, submitted \$1.2M+ daily with 99.9%+ market submission acceptance rate for 6+ months since launch.
- Designed and built an agentic Auto-research system for load/generation forecasting, where agents explore freely with simulation and publish immutable candidate models, AI critic selects production output based on realized performance, and successors inherit experiments, evidence, and trajectories, reducing MAE by 17% versus the prior vendor.
- Proposed and led the adoption of Claude as the company's self-serve analysis and automation layer, growing weekly active usage from 0-82% of employees; led delivery of a unified MCP connector exposing internal/external data and APIs that make Claude really useful, with role based access control and audit logging for data governance.
- Grew monthly MCP events 36x in four months by using AI agents to identify user pain points from audit logs, rapidly resolving user blockers, leading cross-department onboarding, and hosting biweekly company-wide AI Office Hours.
- Initiated and led delivery of multiple AI agentic systems, notably: a security scanner that found 6 critical vulnerabilities, including one in a third-party vendor system; a data anomaly and human-error detection system that helped traders proactively avoid \$100K+ in losses; an RFP finder that helped Commercial surface a real opportunity in its first month.
- Empowered all three team members to achieve 6x traditional engineering productivity by building an AI-native, self-driven, research-to-production culture: encouraging end-to-end ownership, close end-user engagement, engineer-led innovation; hosting weekly AI research seminars; and automating code review, testing, and CI/CD.

Software Engineer

Oct 2024 – Dec 2025

- Architected and built the platform foundation from scratch using Python/GCP, React/TypeScript, and BigQuery; team-owned data pipelines span 1.5B+ production rows; delivering core apps including trading submissions, AI data marketplace, position, settlement & risk management, resource planning map, ticket & notification system.

Elseware Capital | Full Stack Developer (Remote)

Jan 2024 – Sep 2024

- Built from scratch a full-stack real estate trading Web3 app using React, FastAPI, and MySQL, supporting asset tokenization, KYC, portfolio, and live blockchain transactions; optimized APIs and database to reduce latency by 80%.

Ecesis Investment | Frontend Developer Intern (Part-time, Remote)

Jul 2023 – Dec 2023

- Built a cross-platform AI scribing mobile app in React Native with real-time audio transcription and LLM summarization; optimized WebSocket streaming and Redux caching, reducing transcription latency 60% and UI rendering time 80%.

EDUCATION

Cornell University | M.Eng., Computer Engineering | GPA: 3.95/4.00

Aug 2022 – Dec 2023

University of Liverpool | B.Eng. (Honors), Electrical & Electronic Engineering | GPA: 3.97/4.00

Sep 2017 – Jul 2021

Coursework: Deep Learning (A+), Machine Learning (A), Computer Vision (A+), Distributed Computing Principles.

SELECTED PROJECTS

Transformer & Reinforcement Learning [\[Demo\]](#) Built a Transformer policy using supervised pretraining and PPO for predator-prey simulation; distilled the policy by 40% for <10 ms browser inference.

Distributed Systems [\[Code\]](#) Implemented a sharded Java key/value store using Multi-Paxos and two-phase commit for strongly consistent cross-shard transactions.

TECHNICAL SKILLS

AI & Machine Learning: Python, PyTorch, LLM Evaluation, Agentic AI, RAG, MCP, SFT, RL, Speech-to-Text (STT), SQL

Systems & Engineering: TypeScript, React, FastAPI, REST APIs, MySQL, OAuth, RBAC, GCP, AWS, K8s, Docker, CI/CD, Git